

胡志明

个人简历

人本智能实验室
香港科技大学（广州），中国
☎ 020-88333707
✉ zhiminghu@hkust-gz.edu.cn
🌐 zhiminghu.net
博士

个人简介

胡志明博士现为香港科技大学（广州）助理教授、博士生导师，负责领导人本智能实验室。他也是香港科技大学合聘助理教授。他曾于 2022 年 8 月至 2025 年 7 月在德国斯图加特大学进行博士后研究，合作导师为 Andreas Bulling 教授与 Syn Schmitt 教授。他于 2022 年在北京大学获得博士学位，专业为计算机软件与理论，导师为汪国平教授。2017 年，他从北京理工大学获得学士学位，专业为光电信息科学与工程。他的研究兴趣涵盖虚拟现实、增强现实、人机交互、眼动追踪、具身智能、以及人本智能。他在虚拟现实/增强现实、人机交互、以及人工智能领域的顶级期刊和会议上发表了 20 多篇论文，包括 SIGGRAPH、TVCG、IEEE VR、ISMAR、CHI、UIST、以及 AAI。他的研究成果荣获 **ISMAR 2024 最佳期刊论文**、**IEEE VR 2021 最佳期刊论文提名** 以及 **INTERACT 2023 最佳学生论文提名**。他目前是 **TVCG 期刊编委**。他还担任 SIGGRAPH、SIGGRAPH Asia、TVCG、IEEE VR、ISMAR、CHI、UIST、IMWUT、CVPR、ICCV、ECCV、AAI、TMM、IJHCI、以及 TCSVT 等众多顶级期刊和会议的审稿人。

研究方向

胡志明博士的研究兴趣涵盖虚拟现实、增强现实、人机交互、眼动追踪、具身智能、以及人本智能。他主要研究人机交互系统中人类行为的分析与建模，旨在理解人类行为模式，并构建以人为中心的智能交互系统。

学术经历

- 合聘助理教授、博士生导师 2025.10 -
香港科技大学
- 助理教授、博士生导师 2025.08 -
香港科技大学（广州）
- 博士后 2022.08 - 2025.07
斯图加特大学，合作导师：Andreas Bulling 教授，Syn Schmitt 教授

教育背景

- 博士，计算机软件与理论专业 2017.09 - 2022.07
北京大学，导师：汪国平教授
- 学士，光电信息科学与工程专业 2013.09 - 2017.07
北京理工大学

荣誉奖励

- ISMAR 2024 最佳期刊论文
- 德国巴登-符腾堡州基金会博士后研究学者, 2024
- INTERACT 2023 最佳学生论文提名
- SimTech 博士后研究学者, 2022
- 国家奖学金 (前 2%), 2021
- IEEE VR 2021 最佳期刊论文提名
- 国家留学基金委奖学金, 2020
- 校长奖学金 (前 2%), 2020
- 廖凯原奖学金 (前 5%), 2019
- 领航奖学金 (前 0.2%, 7/3800), 2017
- 国家奖学金 (前 2%), 2016
- 国家奖学金 (前 2%), 2014

科研项目

- 混合现实中复杂任务的眼动行为和视觉注意机制与眼动预测, 国家自然科学基金委员会面上项目, 2022.01 至 2025.12, 61 万元, 参与 (排名第三)
- 面向可穿戴设备的机器学习算法研究, 德国研究基金委员会卓越战略项目, 2022.08 至 2024.07, 20 万欧元, 参与
- 智能辅助设备算法设计, 德国巴登-符腾堡州研究基金委员会项目, 2024.08 至 2025.07, 10 万欧元, 参与

学术活动

论文审稿

- 期刊: TVCG, IMWUT, TMM, IJHCI, TCSVT, TiiS, MTAP, VR, BRM
- 会议: SIGGRAPH, SIGGRAPH Asia, CVPR, ICCV, ECCV, CHI, UIST, IEEE VR, ISMAR, AAAI, PG, ETRA

学术组织

- AAAI 2027 程序委员会成员
- ICXR 2026 程序联合副主席
- TVCG 编委
- CCF 虚拟现实与可视化技术专委会执行委员
- AAAI 2026 程序委员会成员
- ETRA 2025 展示与海报主席 (Presentation and Poster Chair)
- AAAI 2025 程序委员会成员
- MuC 2024 副主席 (Associate Chair)
- PETMEI 2024 程序委员会成员
- ETRA 2024 虚拟化主席 (Virtualization Chair)
- MuC 2023 副主席 (Associate Chair)
- iWOAR 2023 程序委员会成员

学术讲座

- 虚拟/增强现实环境中用户眼睛运动和身体运动的协调性, GAMES Webinar 2025, 主持人: 刘鑫达博士, 2025.07
- 人类日常行为中的眼动身体运动协调性以及基于身体运动的眼动预测, ISMAR 2024, 2024.10
- 注视引导的人体运动预测, IROS 2024 非语言的人机智能协作研讨会, 主持人: Jouh Yeong Chew 博士, 2024.10
- 以用户为中心的人工智能, 南京大学第 11 届诚耀青年学者论坛, 2023.12
- 用户感知智能交互系统, 北京大学计算机学院第五届青年论坛, 2023.12
- 眼动、身体运动、与场景的协调性研究, 北京理工大学第十届“特立论坛”, 主持人: 王国仁教授, 2023.11
- 数字人姿态协调性研究, 北京大学计算机学院就业学术讲座, 2022.11
- 虚拟现实环境中用户视觉注意的分析与预测, 东南大学, 主持人: 丁珂教授, 2022.06
- 沉浸式虚拟现实环境中基于眼动和头动信息的用户任务识别, IEEE VR 2022, 主持人: Kiyoshi Kiyokawa 教授, 2022.03
- 任务驱动虚拟现实场景中的用户注视预测, GAMES Webinar 2021, 主持人: 杨旭波教授, 2021.09
- 虚拟现实环境中用户视觉注意的分析与预测, ChinaVR 2020 - IEEE VR 之夜主题论坛, 主持人: 王莉莉教授, 2020.09
- 基于眼动头动协调性的注视预测模型, 2019 国际 VR/AR 暨三维显示大会, 主持人: 徐枫教授, 2019.06

教学经历

- 人本智能, 香港科技大学 (广州), 2026 -
- 机器感知与学习, 斯图加特大学, 2022 - 2025

发表文章

* 通讯作者 # 共同一作

期刊论文

1. Takumi Nishiyasu, **Zhiming Hu**, Andreas Bulling, Yoichi Sato. Learning Alignments of Human Gaze and Fine-grained Task Descriptions. Proceedings of the ACM on Computer Graphics and Interactive Techniques, 2026, 9(2): 1-18.
2. Chuhan Jiao, Yao Wang, Guanhua Zhang, Mihai Băce, **Zhiming Hu**, Andreas Bulling. DiffGaze: A Diffusion Model for Modelling Fine-grained Human Gaze Behaviour on 360° Images. ACM Transactions on Interactive Intelligent Systems, 2026, 16(1): 1-23.
3. **Zhiming Hu***, Guanhua Zhang, Zheming Yin, Daniel Haeufle, Syn Schmitt, Andreas Bulling. HaHeAE: Learning Generalisable Joint Representations of Human Hand and Head Movements in Extended Reality. IEEE Transactions on Visualization and Computer Graphics (oral presentation at ISMAR 2025), 2025, 31(10): 8726 - 8737. (CCF A)
4. **Zhiming Hu***, Zheming Yin, Daniel Haeufle, Syn Schmitt, Andreas Bulling. HOIMotion:

- Forecasting Human Motion During Human-Object Interactions Using Egocentric 3D Object Bounding Boxes. *IEEE Transactions on Visualization and Computer Graphics (ISMAR 2024 Journal-track)*, 2024, 30(11): 7375 - 7385. (**CCF A**, 最佳期刊论文 (会议唯一))
5. **Zhiming Hu***, Jiahui Xu, Syn Schmitt, Andreas Bulling. Pose2Gaze: Eye-body Coordination during Daily Activities for Gaze Prediction from Full-body Poses. *IEEE Transactions on Visualization and Computer Graphics* (oral presentation at ISMAR 2024), 2025, 31(9): 4655-4666. (**CCF A**)
 6. Yao Wang, Yue Jiang, **Zhiming Hu**, Constantin Ruhdorfer, Mihai Băce, Andreas Bulling. VisRecall++: Analysing and Predicting Visualisation Recallability from Gaze Behaviour. *Proceedings of the ACM on Human-Computer Interaction*, 2024, 8(ETRA): 1-18.
 7. Mayar Elfares, Pascal Reiser, **Zhiming Hu**, Wenwu Tang, Ralf Küsters, Andreas Bulling. PrivatEyes: Appearance-based Gaze Estimation Using Federated Secure Multi-Party Computation. *Proceedings of the ACM on Human-Computer Interaction*, 2024, 8(ETRA): 1-23.
 8. Zehui Lin, Xiang Gu, Sheng Li, **Zhiming Hu**, Guoping Wang. Intentional Head-Motion Assisted Locomotion for Reducing Cybersickness. *IEEE Transactions on Visualization and Computer Graphics*, 2023, 29(8): 3458-3471. (**CCF A**)
 9. **Zhiming Hu**, Andreas Bulling, Sheng Li, Guoping Wang. EHTask: Recognizing User Tasks from Eye and Head Movements in Immersive Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics*, 2023, 29(4): 1992-2004. (**CCF A**)
 10. **Zhiming Hu**, Sheng Li, Meng Gai. Research progress of user task prediction and algorithm analysis (in Chinese). *Journal of Graphics*, 2021, 42(3): 367-375.
 11. **Zhiming Hu**, Andreas Bulling, Sheng Li, Guoping Wang. FixationNet: Forecasting Eye Fixations in Task-Oriented Virtual Environments. *IEEE Transactions on Visualization and Computer Graphics (IEEE VR 2021 Journal-track)*, 2021, 27(5): 2681-2690. (**CCF A**, 最佳期刊论文提名 (国内首次))
 12. **Zhiming Hu**, Sheng Li, Congyi Zhang, Kangrui Yi, Guoping Wang, Dinesh Manocha. DGaze: CNN-Based Gaze Prediction in Dynamic Scenes. *IEEE Transactions on Visualization and Computer Graphics (IEEE VR 2020 Journal-track)*, 2020, 26(5): 1902-1911. (**CCF A**)
 13. **Zhiming Hu**, Sheng Li, Meng Gai. Temporal continuity of visual attention for future gaze prediction in immersive virtual reality. *Virtual Reality and Intelligent Hardware*, 2020, 2(2): 142-152.
 14. **Zhiming Hu**, Congyi Zhang, Sheng Li, Guoping Wang, Dinesh Manocha. SGaze: A Data-Driven Eye-Head Coordination Model for Realtime Gaze Prediction. *IEEE Transactions on Visualization and Computer Graphics (IEEE VR 2019 Journal-track)*, 2019, 25(5): 2002-2010. (**CCF A**)

会议论文

1. Chuhan Jiao, Guanhua Zhang, Yeonjoo Cho, **Zhiming Hu**, Andreas Bulling. DiffEyeSyn: User-specific Subtle Eye Movement Synthesis Using Diffusion Models. *Proceedings of the IEEE International Conference on Automatic Face and Gesture Recognition*, 2026.

2. Qing Chang#, **Zhiming Hu**#. GazeInterpreter: Parsing Eye Gaze to Generate Eye-Body-Coordinated Narrations. Proceedings of the AAAI Conference on Artificial Intelligence, 2026, 40(21): 17410-17418. (CCF A)
3. Chuhan Jiao, **Zhiming Hu***, Andreas Bulling. HAGI: Head-Assisted Gaze Imputation for Mobile Eye Trackers. Proceedings of the ACM Symposium on User Interface Software and Technology, 2025: 1-14. (CCF A)
4. **Zhiming Hu***, Daniel Haeufle, Syn Schmitt, Andreas Bulling. HOIGaze: Gaze Estimation During Hand-Object Interactions in Extended Reality Exploiting Eye-Hand-Head Coordination. Proceedings of the ACM Special Interest Group on Computer Graphics and Interactive Techniques, 2025: 1-10. (CCF A)
5. Guanhua Zhang, Mohamed Ahmed, **Zhiming Hu***, Andreas Bulling. SummAct: Uncovering User Intentions Through Interactive Behaviour Summarisation. Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, 2025: 1-17. (CCF A)
6. Haodong Yan#, **Zhiming Hu**#, Syn Schmitt, Andreas Bulling. GazeMoDiff: Gaze-guided Diffusion Model for Stochastic Human Motion Prediction. Proceedings of the Pacific Conference on Computer Graphics and Applications, 2024: 1-12.
7. Guanhua Zhang, **Zhiming Hu***, Andreas Bulling. DisMouse: Disentangling Information from Mouse Movement Data via Diffusion Models. Proceedings of the ACM Symposium on User Interface Software and Technology, 2024: 1-13. (CCF A)
8. **Zhiming Hu***, Syn Schmitt, Daniel Haeufle, Andreas Bulling. GazeMotion: Gaze-guided Human Motion Forecasting. Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems, 2024: 13017-13022. (口头报告)
9. Guanhua Zhang, **Zhiming Hu***, Mihai Băce, Andreas Bulling. Mouse2Vec: Learning Reusable Semantic Representations of Mouse Behaviour. Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, 2024: 1-17. (CCF A)
10. Yao Wang, Weitian Wang, Abdullah Abdelhafez, Mayar Elfares, **Zhiming Hu***, Mihai Băce, Andreas Bulling. SalChartQA: Question-driven Saliency on Information Visualisations. Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, 2024: 1-14. (CCF A)
11. Chuhan Jiao, **Zhiming Hu***, Mihai Băce, Andreas Bulling. SUPREYES: SUPER Resolution for EYES Using Implicit Neural Representation Learning. Proceedings of the ACM Symposium on User Interface Software and Technology, 2023: 1-13. (CCF A)
12. Guanhua Zhang, Matteo Bortoletto, **Zhiming Hu***, Lei Shi, Mihai Băce, Andreas Bulling. Exploring Natural Language Processing Methods for Interactive Behaviour Modelling. Proceedings of the IFIP Conference on Human-Computer Interaction, 2023: 3-26. (最佳学生论文提名)

摘要与短文

1. Yi Zou, Ziming Li, Hai-Ning Liang*, **Zhiming Hu***. Contextual Recovery: Guiding Hand Tracking Failures Recovery in Mixed Reality via VLM Reasoning. Proceedings of the 2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops, 2026:

368-372.

2. Mayar Elfares, **Zhiming Hu**, Pascal Reisert, Andreas Bulling, Ralf Küsters. Federated Learning for Appearance-based Gaze Estimation in the Wild. Proceedings of the NeurIPS Workshop Gaze Meets ML, 2023: 20-36.
3. **Zhiming Hu**. Eye Fixation Forecasting in Task-Oriented Virtual Reality. Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops, 2021: 707-708.
4. **Zhiming Hu**. Gaze Analysis and Prediction in Virtual Reality. Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops, 2020: 543-544.